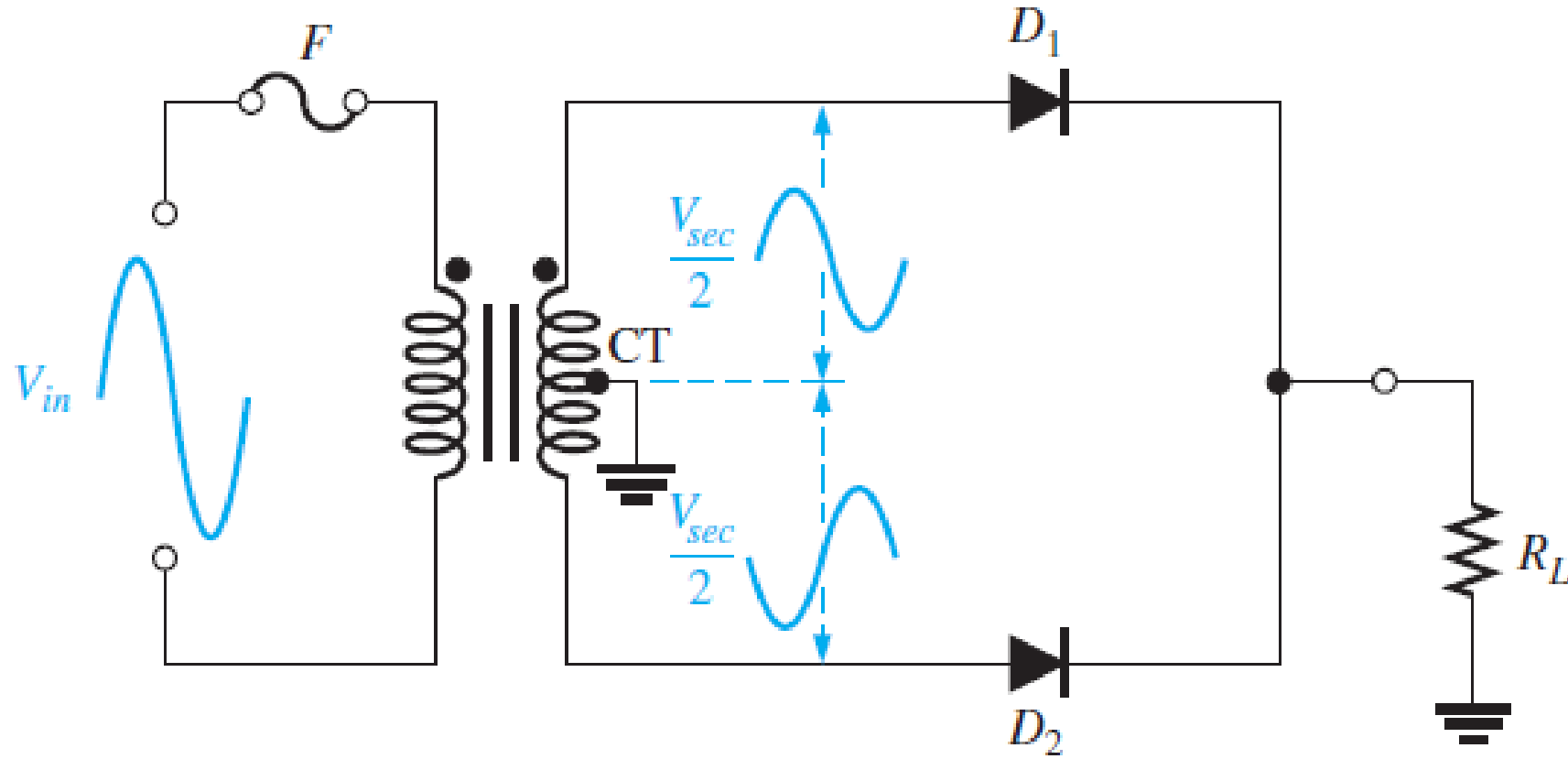


Full Wave Rectifier

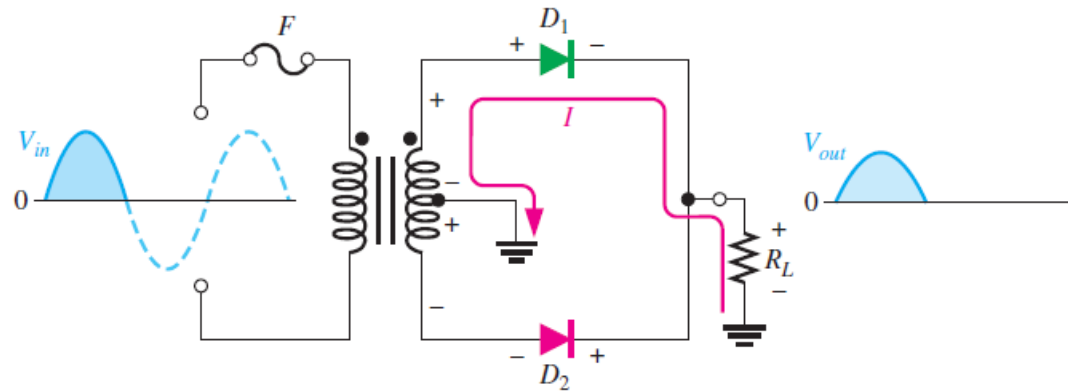
- Full wave rectifier allows unidirectional current for entire 360° of input cycle
- It is combination of two half wave rectifiers
- For this purpose two diodes are used with a centre taped transformer which provides two separate voltages (out of phase) across its secondary winding
- One diode conducts and other diode is reverse biased during positive input cycle and vice versa
- As a result current is continuously flow through the load resistor

Full Wave Rectifier (contd.)

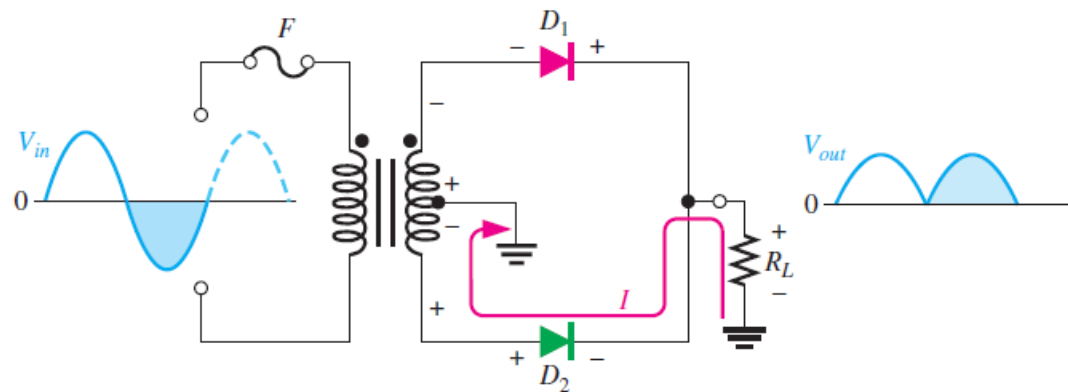


Full Wave Rectifier (contd.)

Basic operation of a center-tapped full-wave rectifier. Note that the current through the load resistor is in the same direction during the entire input cycle, so the output voltage always has the same polarity.



(a) During positive half-cycles, D_1 is forward-biased and D_2 is reverse-biased.



(b) During negative half-cycles, D_2 is forward-biased and D_1 is reverse-biased.

Full Wave Rectifier (contd.)

- The average value of full wave rectifier can be calculated as:

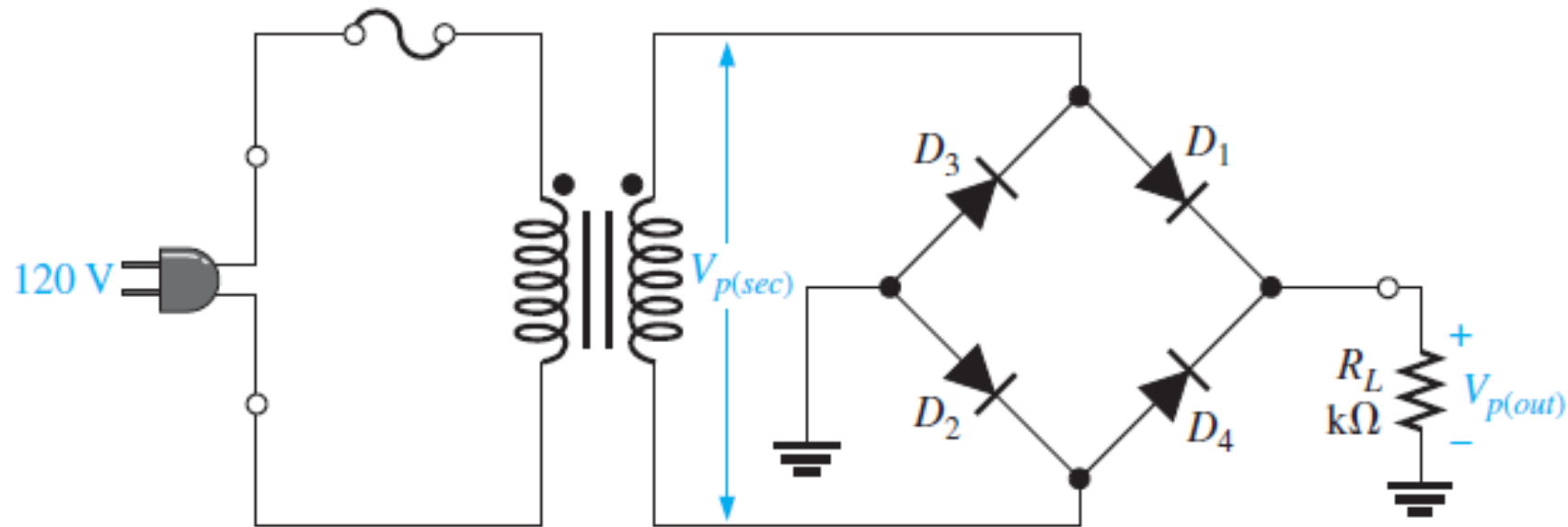
$$V_{Avg} = \frac{2V_P}{\pi}$$

- Equation shows that V_{avg} is approx. 63.6% of V_p (Ideal Diode Case)
- The frequency of full wave rectifier will be equal to

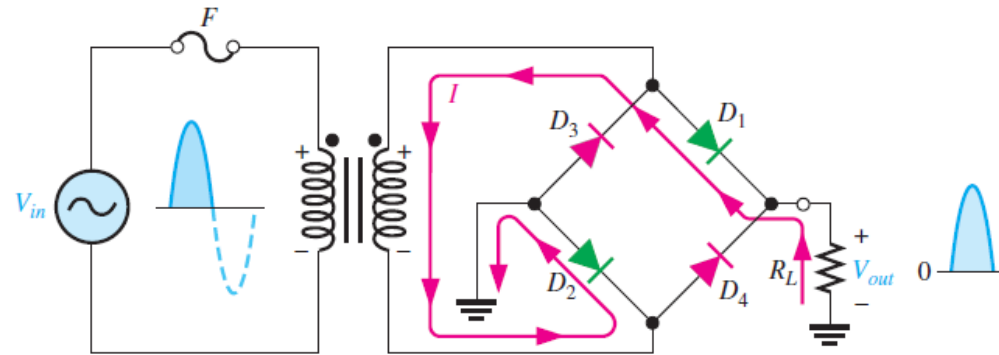
$$f_{out} = 2f_{in}$$

Full Wave Rectifier (contd.)

- Another type of full wave rectifier is Bridge Rectifier

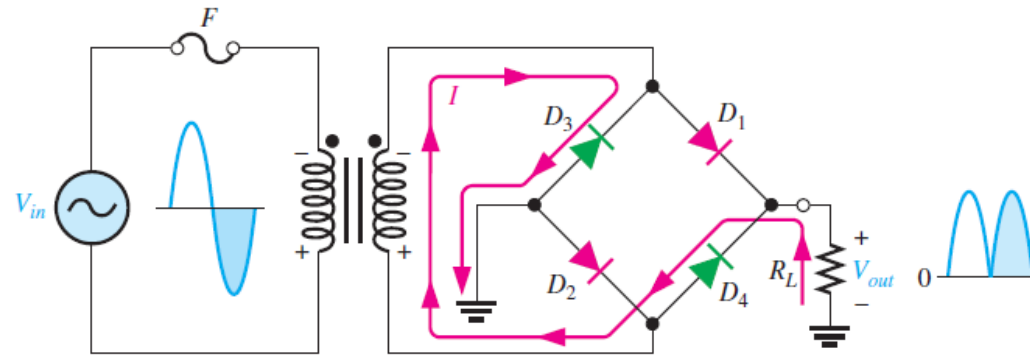


Full Wave Rectifier (contd.)



Operation of a bridge rectifier.

(a) During the positive half-cycle of the input, D_1 and D_2 are forward-biased and conduct current. D_3 and D_4 are reverse-biased.



(b) During the negative half-cycle of the input, D_3 and D_4 are forward-biased and conduct current. D_1 and D_2 are reverse-biased.

Full Wave Rectifier (contd.)

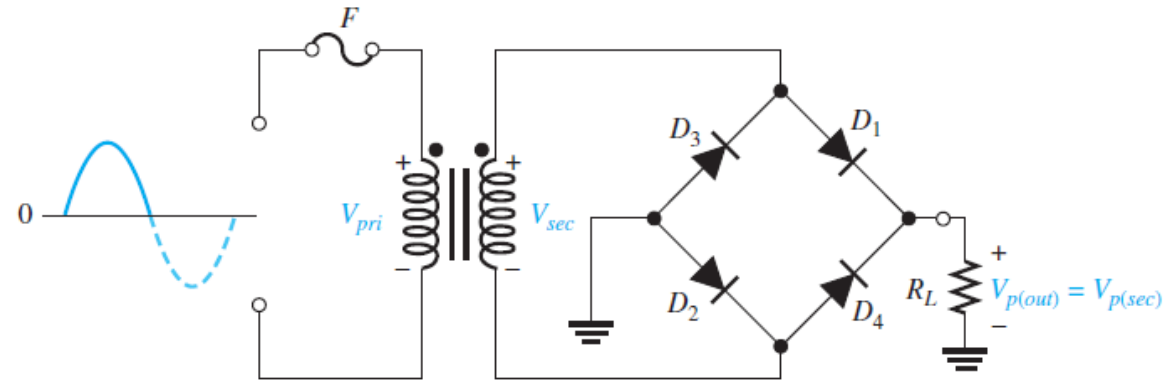
- the bridge output voltage in case of ideal approx. can be calculated as:

$$V_{P(out)} = V_{P(sec)}$$

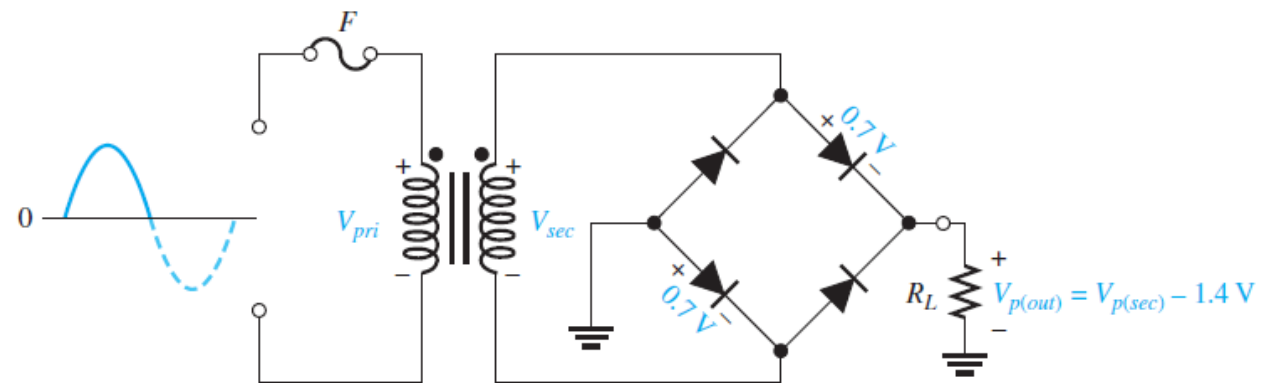
- By using the second approx. the bridge output voltage can be calculated as:

$$V_{P(out)} = V_{P(sec)} - 1.4V$$

Full Wave Rectifier (contd.)



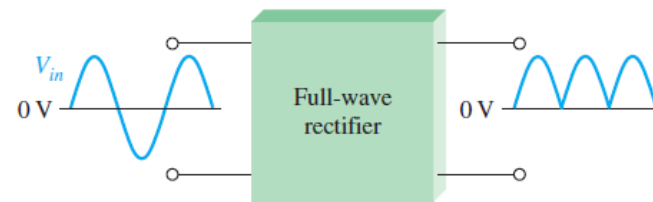
(a) Ideal diodes



(b) Practical diodes (Diode drops included)

Power Supply Filtering

- For a power supply there must be a constant voltage amplitude without fluctuations
- The output of a Full wave or Half Wave rectifier is not constant
- There must be some filtering to smoothen the output of rectifiers



(a) Rectifier without a filter

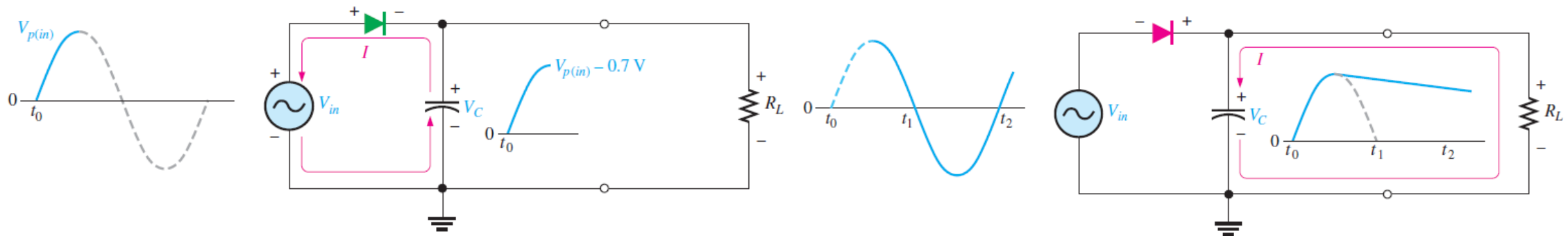


(b) Rectifier with a filter (output ripple is exaggerated)

Power Supply Filtering (contd.)

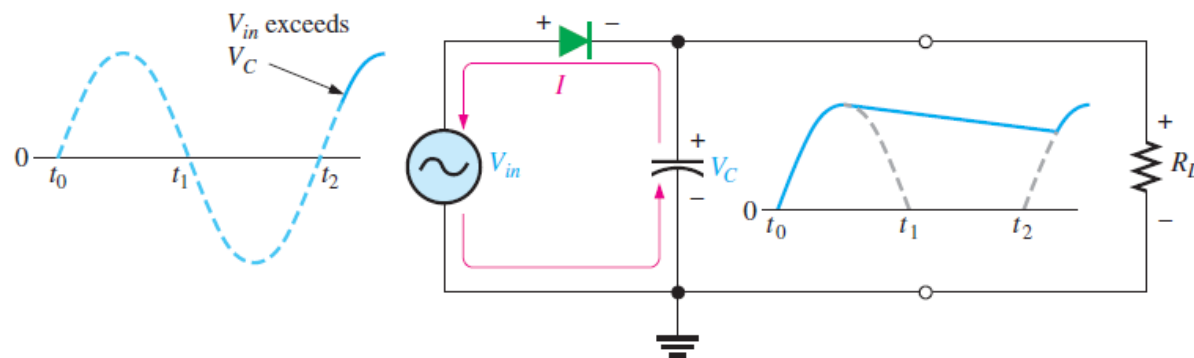
- Capacitor input filter is used for filtering
- Capacitor is attached at the output of rectifier
- when the positive cycle arrived, diode becomes forward bias
- The capacitor start charging and it continues as voltage is increasing, when voltage starts decreasing, capacitor starts discharging and diode becomes reverse bias
- The time constant RC determines the discharging rate of capacitor
- Larger the time constant, lesser the capacitor discharge

Power Supply Filtering (contd.)



(a) Initial charging of the capacitor (diode is forward-biased) happens only once when power is turned on.

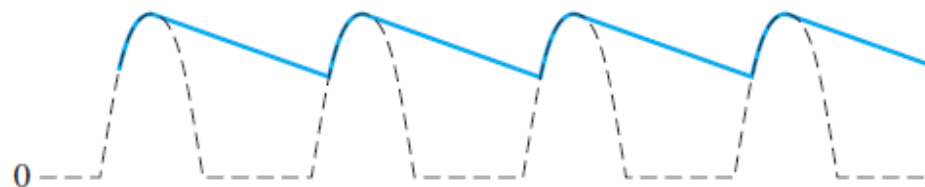
(b) The capacitor discharges through R_L after peak of positive alternation when the diode is reverse-biased. This discharging occurs during the portion of the input voltage indicated by the solid dark blue curve.



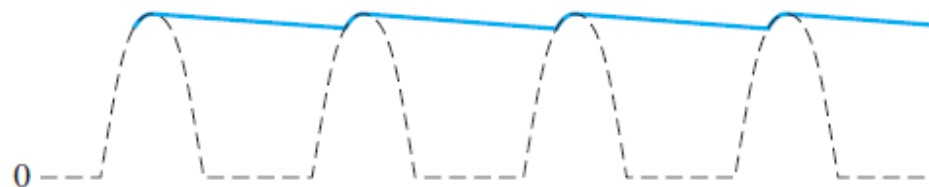
(c) The capacitor charges back to peak of input when the diode becomes forward-biased. This charging occurs during the portion of the input voltage indicated by the solid dark blue curve.

Power Supply Filtering (contd.)

- Capacitor quickly charge and slowly discharge during the complete cycle
- Variation in the capacitor voltage due to charging and discharging is known as ripple voltage
- Smaller the ripple, better the filtering



(a) Larger ripple (blue) means less effective filtering.

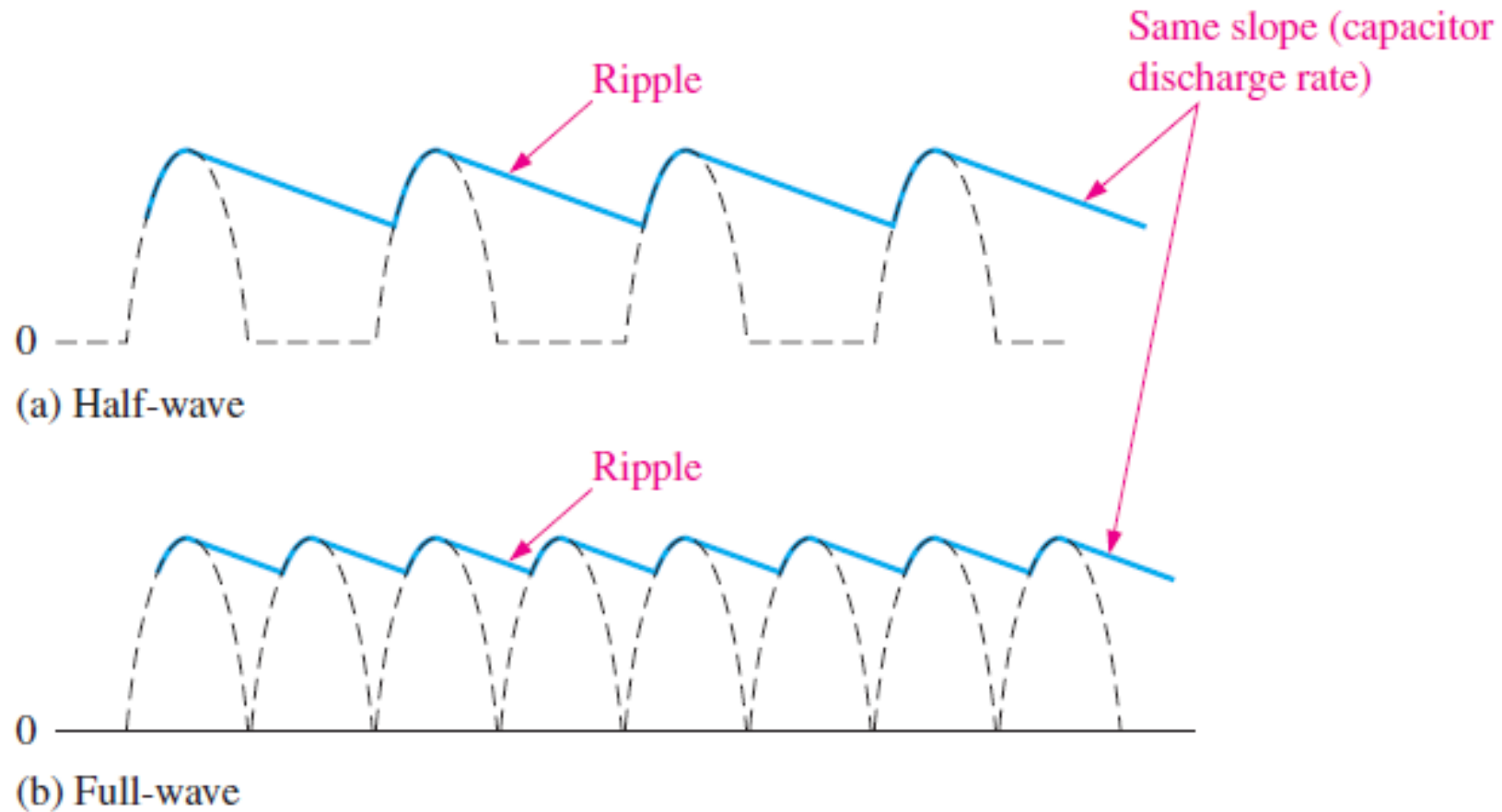


(b) Smaller ripple means more effective filtering. Generally, the larger the capacitor value, the smaller the ripple for the same input and load.

Power Supply Filtering (contd.)

- Full wave rectifier has double the frequency as compare to half wave rectifier
- It is easier to filter the full wave rectifier output as there is short time between peaks
- When filtered with same load resistor and capacitor, full wave rectifier has small ripple as compared to half wave rectifier
- Because capacitor discharges less during the short intervals between full peaks

Power Supply Filtering (contd.)



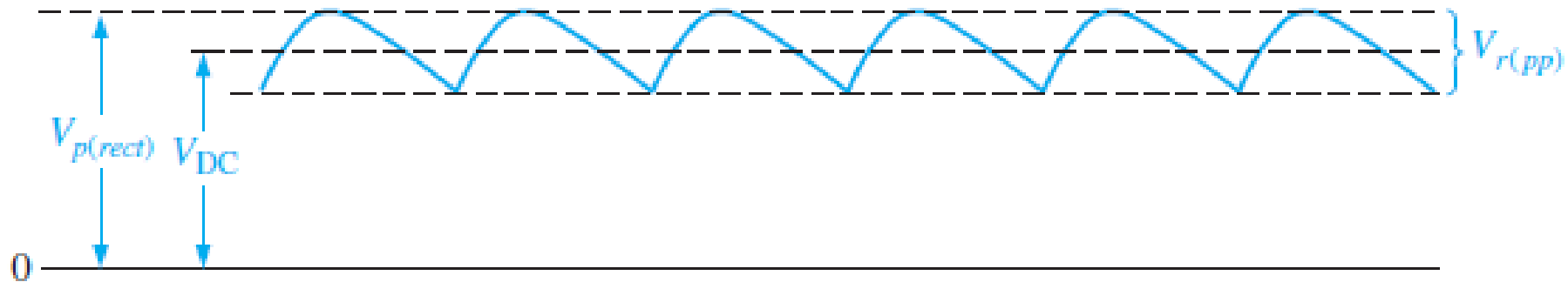
Power Supply Filtering (contd.)

- The ripple factor (r) (amount of AC content present in DC output) is effectiveness of filter and defined as

$$r = \frac{V_r(pp)}{V_{DC}}$$

$$V_{r(pp)} \cong \left(\frac{1}{fR_L C} \right) V_{p(rect)}$$

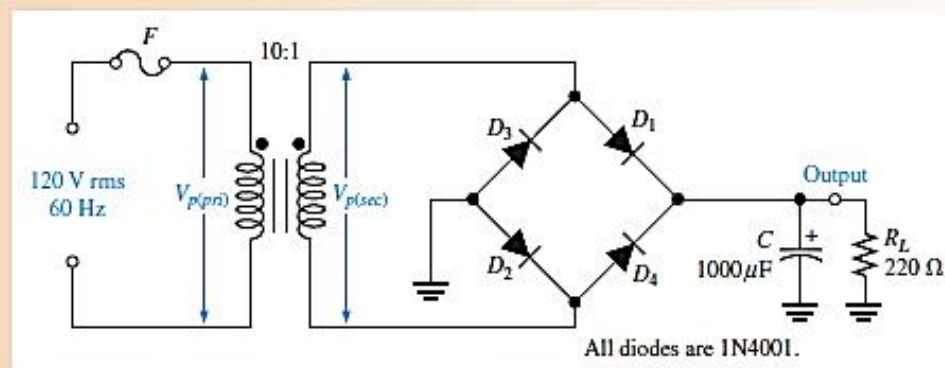
$$V_{DC} \cong \left(1 - \frac{1}{2fR_L C} \right) V_{p(rect)}$$



EXAMPLE XXX

Determine the ripple factor for the filtered bridge rectifier with a load as indicated in Figure XXX.

► **FIGURE XXX**



Solution The transformer turns ratio is $n = 0.1$. The peak primary voltage is

$$V_{p(prim)} = 1.414V_{rms} = 1.414(120\text{ V}) = 170\text{ V}$$

The peak secondary voltage is

$$V_{p(sec)} = nV_{p(prim)} = 0.1(170\text{ V}) = 17.0\text{ V}$$

The unfiltered peak full-wave rectified voltage is

$$V_{p(rect)} = V_{p(sec)} - 1.4\text{ V} = 17.0\text{ V} - 1.4\text{ V} = 15.6\text{ V}$$

The frequency of a full-wave rectified voltage is 120 Hz. The approximate peak-to-peak ripple voltage at the output is

$$V_{r(pp)} \cong \left(\frac{1}{fR_L C} \right) V_{p(rect)} = \left(\frac{1}{(120\text{ Hz})(220\ \Omega)(1000\ \mu\text{F})} \right) 15.6\text{ V} = 0.591\text{ V}$$

The approximate dc value of the output voltage is determined as follows:

$$V_{DC} = \left(1 - \frac{1}{2fR_L C} \right) V_{p(rect)} = \left(1 - \frac{1}{(240\text{ Hz})(220\ \Omega)(1000\ \mu\text{F})} \right) 15.6\text{ V} = 15.3\text{ V}$$

The resulting ripple factor is

$$r = \frac{V_{r(pp)}}{V_{DC}} = \frac{0.591\text{ V}}{15.3\text{ V}} = 0.039$$

The percent ripple is 3.9%.